

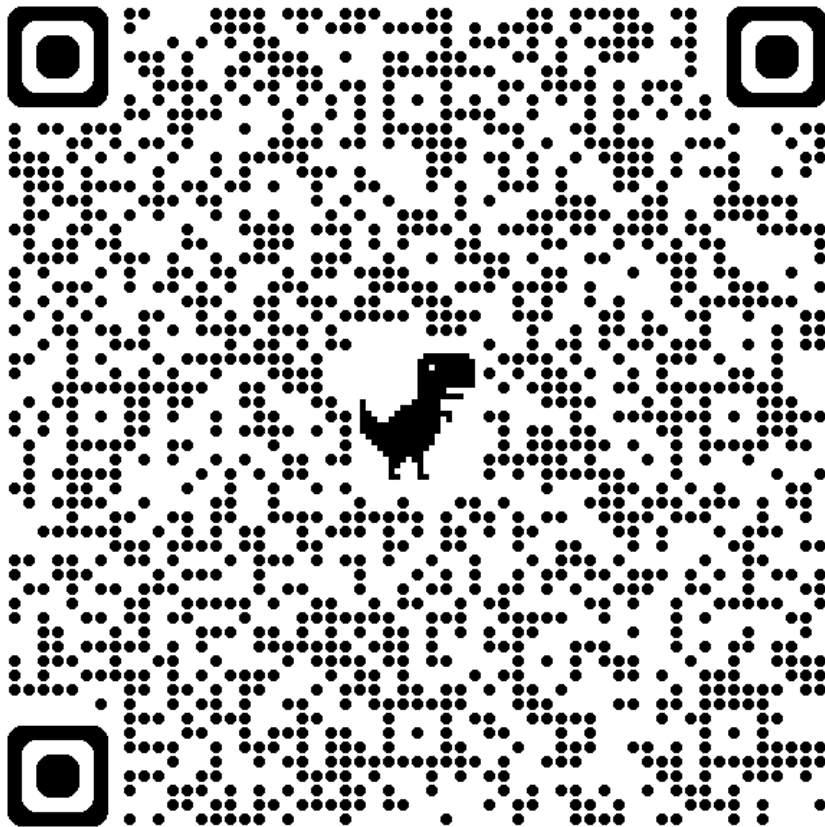


Outline:

- Atomic Densities
 - Linear Density
 - Planar Density
- Single- vs poly- crystalline materials
- X-ray Diffraction
- Example
- Polymorphism and Allotropy

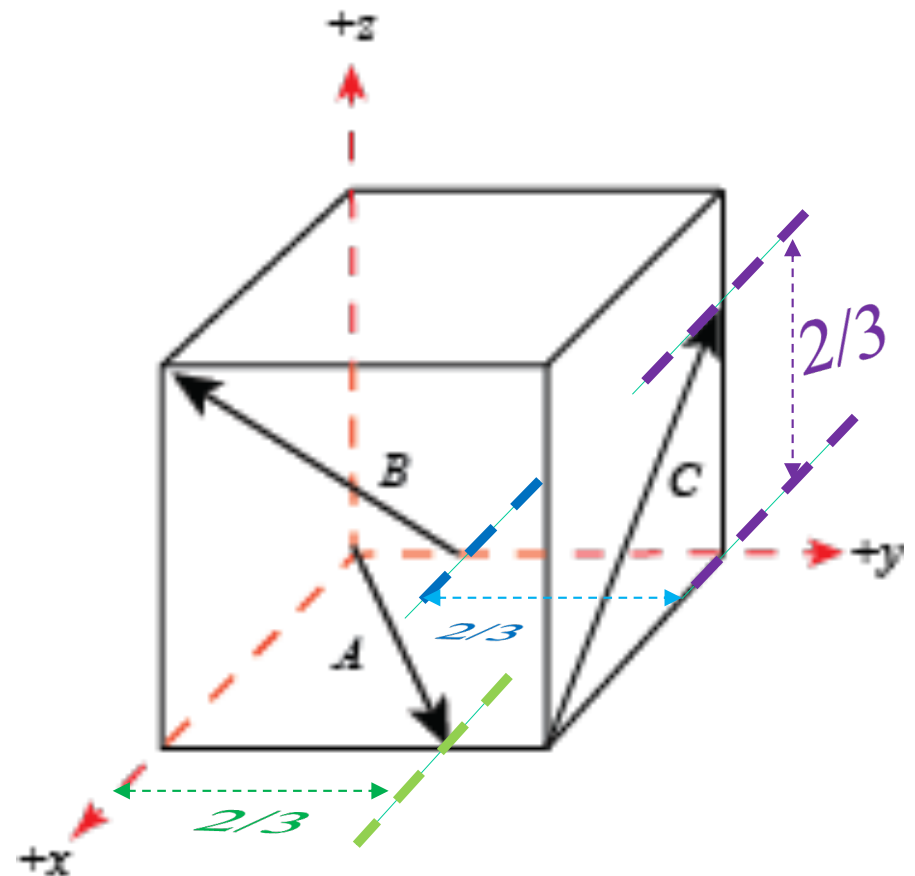


Review - Directions



Use the QR code shown in the classroom. This one won't work.

Which direction is $[\bar{3}\bar{1}3]$





Atomic Densities

- **Linear Density**

- Number of atoms per length whose centers lie on the direction vector for a specific crystallographic direction.

$$\text{Linear Density} = \frac{\text{Number of atoms centered on direction vector}}{\text{length of direction vector}}$$

- **Planar Density**

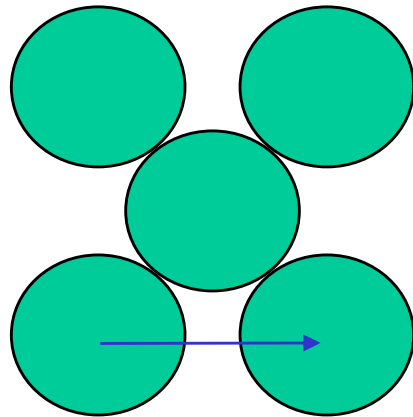
- Number of atoms per unit area that are centered on a particular crystallographic plane.

$$\text{Planar Density} = \frac{\text{Number of atoms centered on a plane}}{\text{area of the plane}}$$



Linear Density

- Calculate the linear density of the [100] direction for the FCC crystal



$$L_D = n/L_L \quad \text{linear density}$$
$$n = 1 \quad \text{atoms}$$
$$L_L = a \quad \text{line length}$$

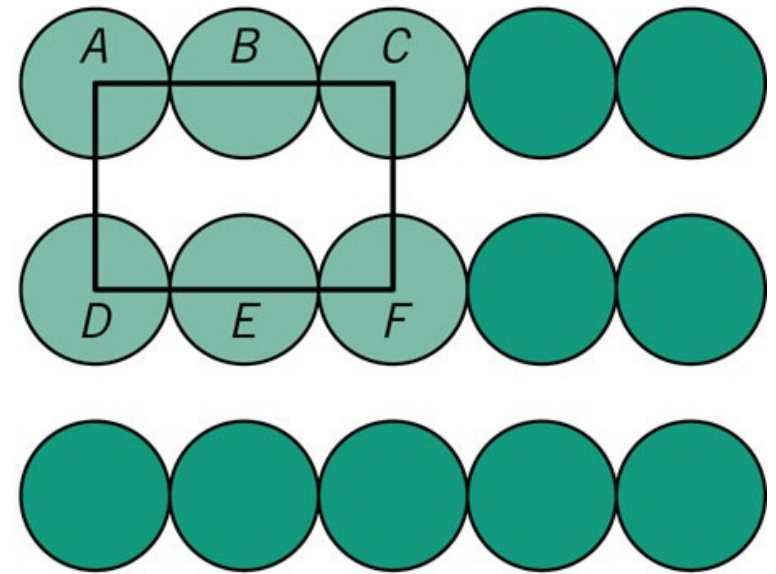
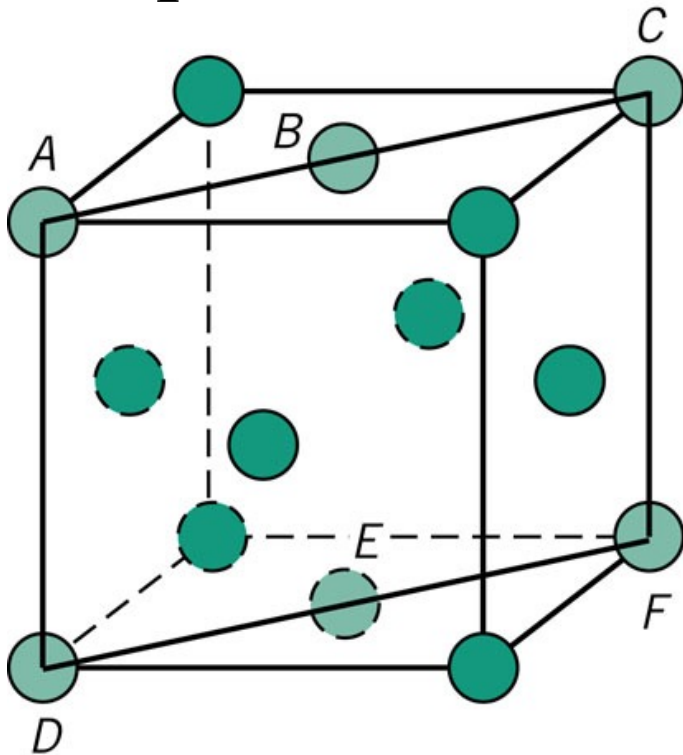
$$\text{For FCC } a = 2 \cdot 2 R$$

$$L_D = .1 / (2 \cdot 2 R)$$



Planar Density

- Calculate the planar density of the (110) plane for the FCC crystal

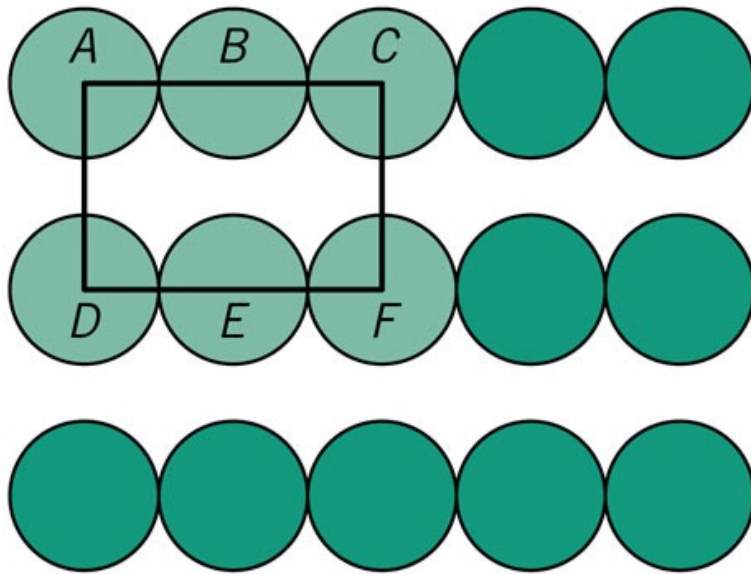


- Compute planar area
- Compute number of atoms

For an atom to be counted, it has to be centered on that plane.



Planar Density



Plane area $A_p = (AC) \times (AD)$

$$AC = 4R$$

$$AD = 2R\sqrt{2}$$

$$A_p = (4R)(2R\sqrt{2}) = 8R^2\sqrt{2}$$

Number of atoms **2 atoms**

$$PD = \frac{n}{A_p} = \underline{\hspace{2cm}}$$



Linear and Planar Density

- Why do we care?
- Properties, in general, depend on linear and planar density.
- Examples:
 - Speed of sound along directions
 - **Slip** ... *in metals*) depends on linear and planar density
 - Slip occurs on planes that have the greatest density of atoms in direction with highest density
 - we would say along closest packed directions on the closest packed planes*



Crystals As Building Blocks

- *Some* engineering applications require **single** crystals:



Single crystal

(Courtesy P.M. Anderson)

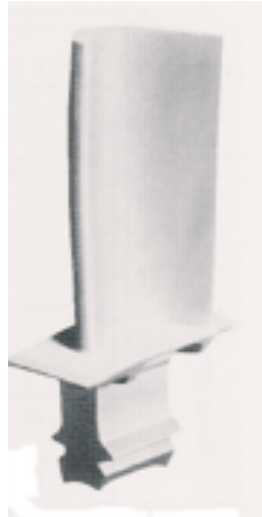


Fig. 8.30(c), *Callister 6e*.
(courtesy of Pratt and Whitney)

- Turbine blades
- Nickel alloy – single crystal
- to improve high temp. mechanical properties



(Courtesy GE Superabrasives)

diamond single crystals
for abrasives

- *Most* engineering materials are **Single crystal**

- Each "grain" is a **single** crystal.
- If crystals are **randomly** oriented,
overall component properties are **not** directional.
- Crystal sizes typically range from 1 nm to 2 cm
(i.e., *from a few to millions of atomic layers*).



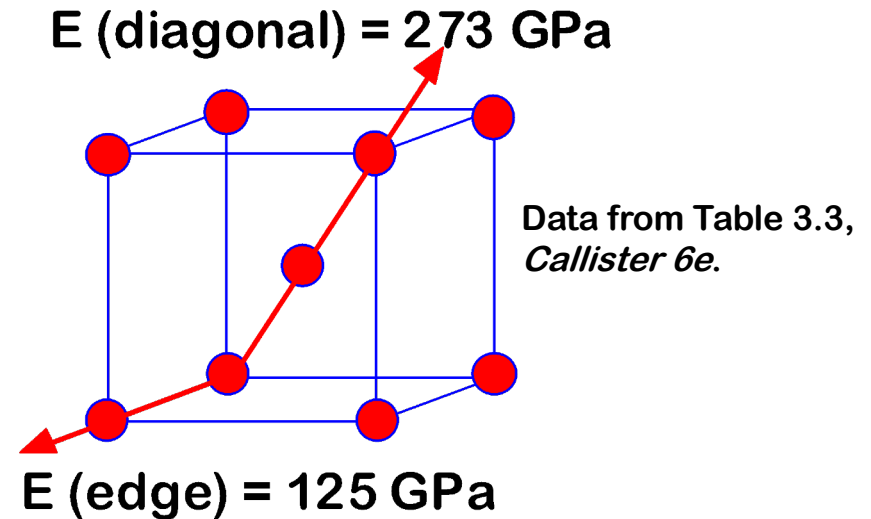
Nb-Hf-W plate with an electron beam weld



Single Vs Polycrystals

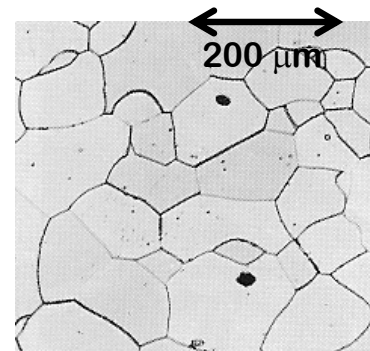
- **Single** Crystals

- Properties vary with direction: **Anisotropic**
- Example: the modulus of elasticity (E) in BCC iron:

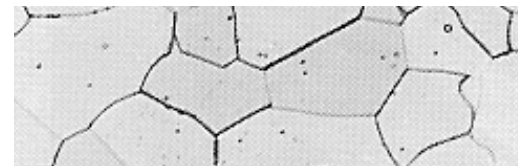


- **Polycrystals**

- Properties may/may not vary with direction.
- If grains are randomly oriented: **Isotropic**
($E_{poly iron} = 210 \text{ GPa}$)
- If grains are **textured**, anisotropic.



Adapted from Fig. 4.12(b), *Callister 6e*.

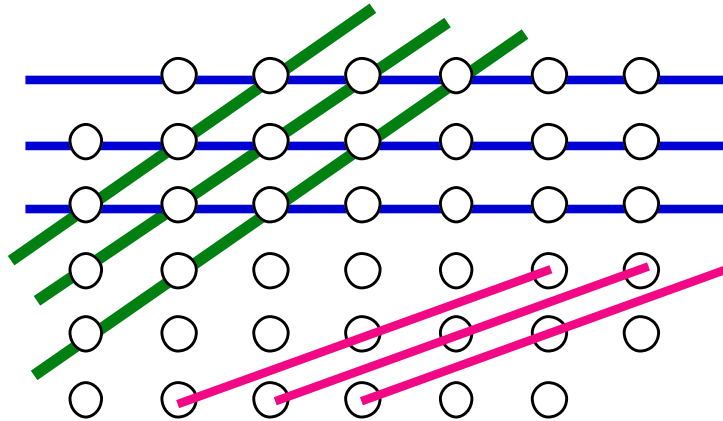


Textured grains



Inter-Planar Spacing & X-Ray Diffraction

2D



- Inter-planar spacing
 - The inter-planar spacing in a **particular direction** is the distance between equivalent planes of atoms
- The existence of planes and the distances between different sets of them in a material are characteristic for each material
 - ➔ Hence, inter-planar spacings are measured by **x-ray diffraction** to identify unknown materials!

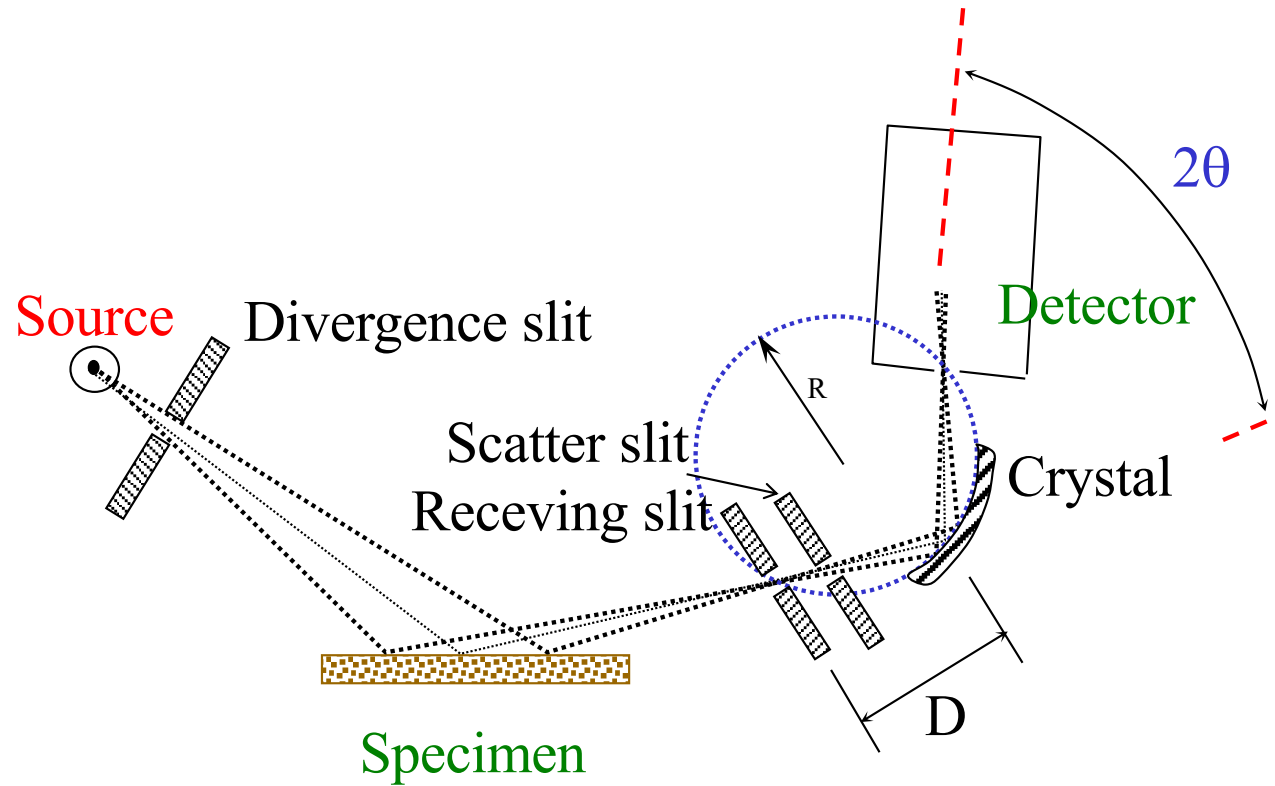


X-Ray Diffraction

- Can be used to determine **crystal structure**
 - identify **unknown** materials
 - measure lattice parameters
- X-rays are a form of electromagnetic radiation that have high energies and short wavelengths.
- Diffraction occurs whenever a wave encounters a series of regularly spaced obstacles that;
 - Can scatter the wave
 - Have a spacing comparable to the **Polymorphism / Allotropy**
- **X-ray wavelength** (λ) $\sim$ inter-atomic spacing.
- Other techniques such as **neutron** or **electron** diffraction, also, can be used.

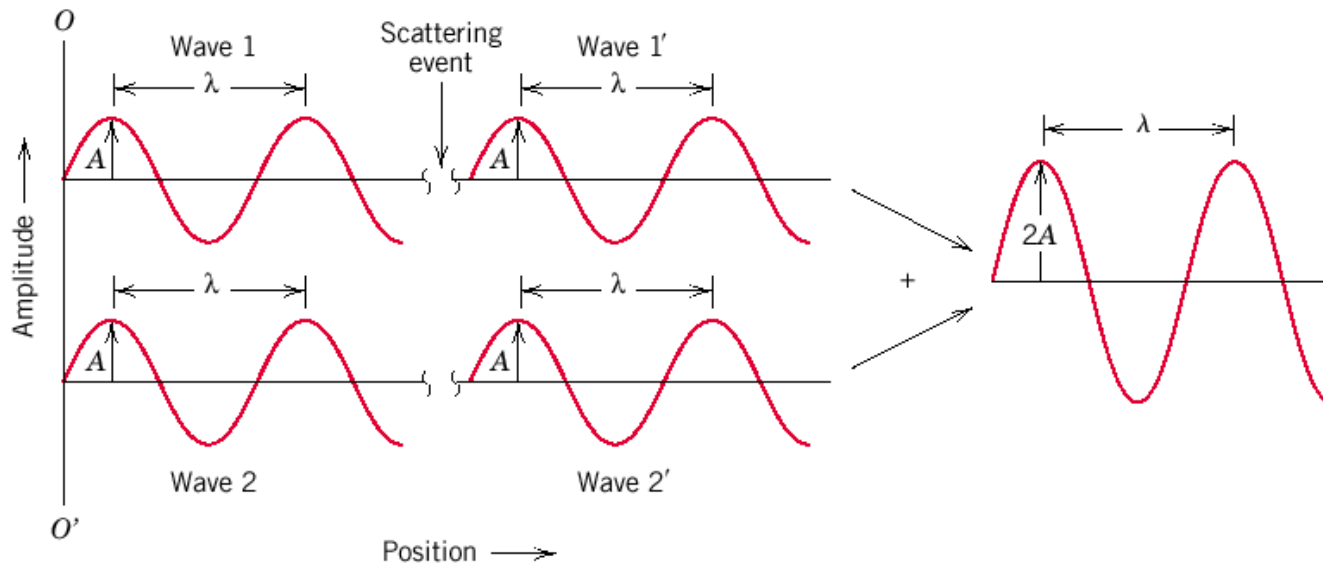


X-ray Diffractometer

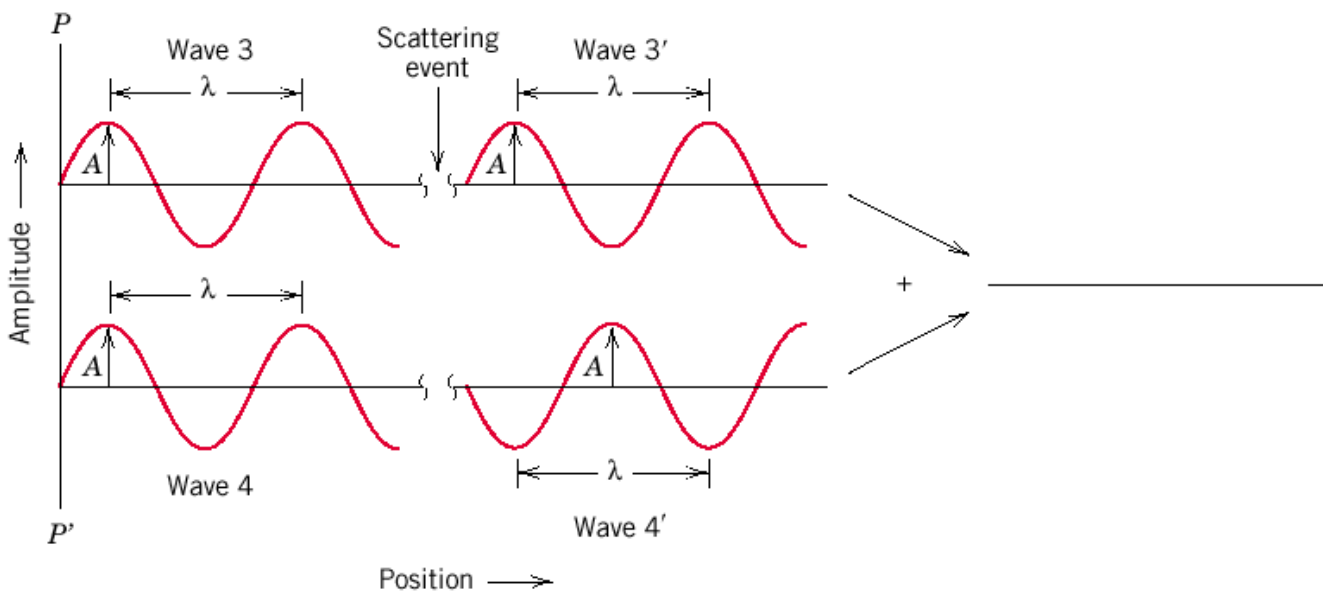




Constructive & Destructive Interference



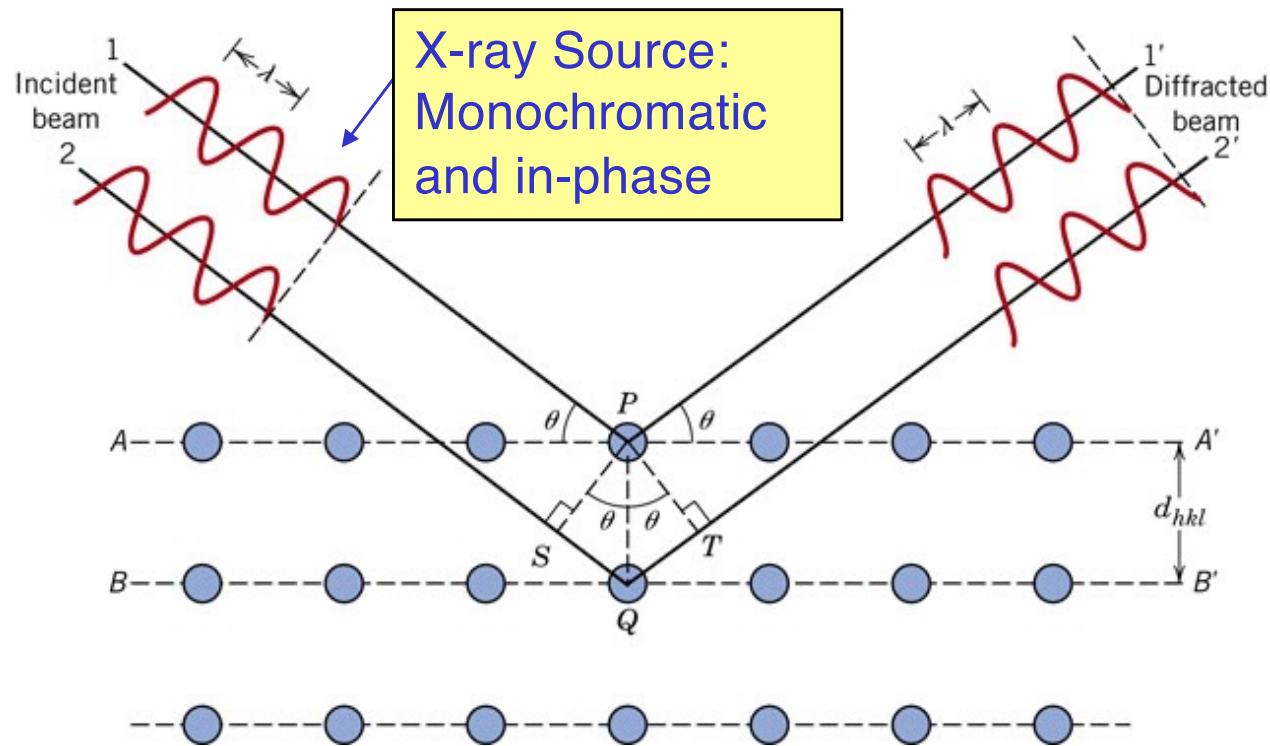
Constructive



Destructive



Bragg's Law



The law: For constructive interference, the additional path length $SQ + QT$ must be an integer number of wavelength (λ).

$$n\lambda = SQ + QT = d_{hkl}\sin\theta + d_{hkl}\sin\theta = 2 d_{hkl}\sin\theta$$

$n = 1, 2, 3 \dots$ order of reflection



Bragg's Law

$$n\lambda = d_{hkl} \sin \theta + d_{hkl} \sin \theta$$

$$= 2d_{hkl} \sin \theta$$

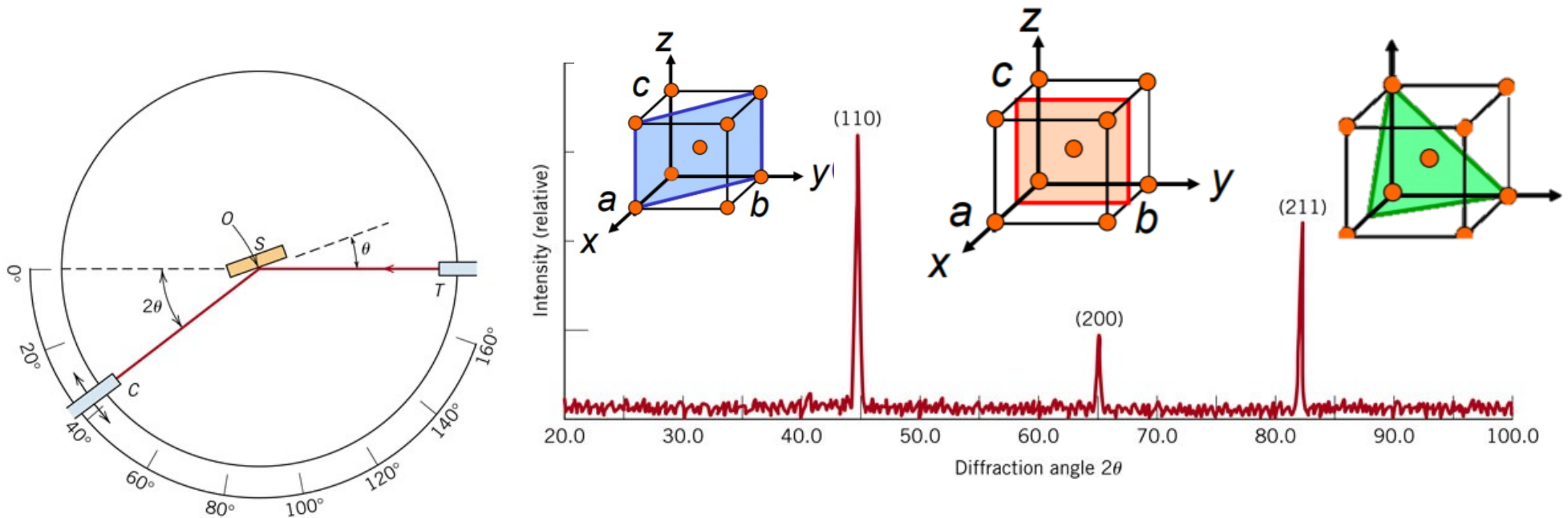
- We have a simple expression relating the **x-ray wavelength** and **interatomic spacing** to the **angle** of the diffracted beam.
- If **Bragg's law** is not satisfied, then the interference will be **nonconstructive** in nature so as to yield a very low-intensity diffracted beam.
- Magnitude of difference between two adjacent and parallel planes of atoms is function of **Miller Indices** and the **lattice parameter**. For cubic symmetry:

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$



Diffractometer Technique

- Use **powder** (or polycrystalline) sample to guarantee some particles will be oriented properly such that every possible set of crystallographic planes will be available for diffraction.



Each material has a unique set of planar distances and extinctions, making X-ray diffraction useful in analysis of an unknown material.



Example

For BCC iron (Fe), compute:

- (a) the **interplanar spacing** of the (220) planes, and
- (b) the **diffraction angle** at which the (220) reflection occurs.
 - Given: *lattice parameter of Fe, $a = 0.2866 \text{ nm}$* ;
 - *X-ray wavelength, $\lambda = 0.1790 \text{ nm}$* ;
 - *first-order reflection ($n = 1$)*.

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

$$\sin \theta = \frac{n\lambda}{2d_{hkl}}$$

$$\theta = 13.7^\circ$$

$$2\theta = 27.4^\circ$$



Polymorphism and Allotropy

- Some materials may have more than one **crystal structure** depending on **temperature** and **pressure** - called *POLYMORPHISM*
- Carbon
 - graphite
 - diamond
- Iron BCC and FCC



Next topic:
Imperfections in Solids